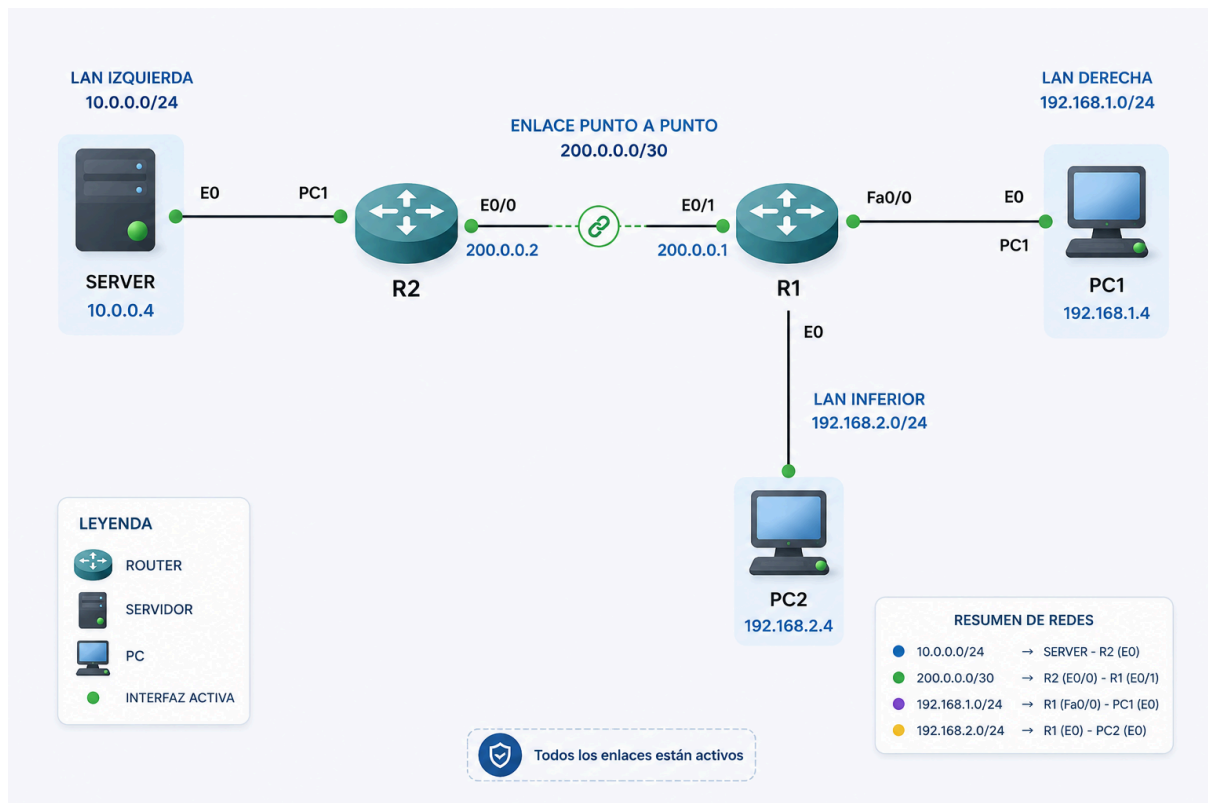


OSPF OPEN SHORTEST PATH FIRST



1) Interface configuration of Router 1, total of 3 interfaces up.

```
inserthostname-here(config)#hostname R1
R1(config)#int eth0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#
*May 20 03:39:46.636: %LINK-5-UPDOWN: Interface Ethernet0/0, changed state to up
*May 20 03:39:47.637: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed state to up
R1(config-if)#int eth0/2
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#int eth0/1
*May 20 03:40:21.669: %LINK-5-UPDOWN: Interface Ethernet0/2, changed state to up
*May 20 03:40:22.668: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/2, changed state to up
R1(config-if)#int eth0/1
R1(config-if)#ip address 200.0.0.1 255.255.255.252
R1(config-if)#no shut
```

2) Interface configuration on router 2, total of 2 interfaces up.

```
inserthostname-here(config)#hostname R2
R2(config)#int eth0/0
R2(config-if)#ip address 200.0.0.2 255.255.255.252
R2(config-if)#no shut
R2(config-if)#int eth0/1
*May 20 03:46:09.653: %LINK-5-UPDOWN: Interface Ethernet0/0, changed state to up
*May 20 03:46:10.654: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed state to up
R2(config-if)#int eth0/1
R2(config-if)#ip address 10.0.0.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#ex
R2(config)#
*May 20 03:46:27.829: %LINK-5-UPDOWN: Interface Ethernet0/1, changed state to up
*May 20 03:46:28.830: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to up
R2(config)#
```

- 3) IP address assignment to pcs (Linux) with default route.

PC1 LAN 192.168.1.0/24

```
inserthostname-here:~$ sudo ip addr add 192.168.1.4/24 dev eth0
inserthostname-here:~$ ip route add default via 192.168.1.1
ip: RTNETLINK answers: Operation not permitted
inserthostname-here:~$ sudo ip route add default via 192.168.1.1
```

- 4) PC2 LAN 192.168.2.0/24

```
inserthostname-here:~$ sudo ip addr add 192.168.2.4/24 dev eth0
inserthostname-here:~$ sudo ip route add default via 192.168.2.1
```

- 5) SVR LAN 10.0.0.0/24

```
inserthostname-here:~$ sudo ip addr add 10.0.0.4/24 dev eth0
inserthostname-here:~$ sudo ip route add default via 10.0.0.1
```

- 6) Activation OSPF on R2, maintain area 0.

```
R2(config)#router ospf 1
R2(config-router)#network
*May 20 03:57:37.197: %OSPF-6-DFT_OPT: Protocol timers for fast convergence are Enabled.
R2(config-router)#network 200.0.0.0 0.0.0.3 area 0
R2(config-router)#network 10.0.0.0 0.0.0.255 area 0
^
% Invalid input detected at '^' marker.
R2(config-router)#network 10.0.0.0 0.0.0.255 area 0
R2(config-router)#router-id 2.2.2.2
R2(config-router)#
R1(config)#router ospf 1
R1(config-router)#e
*May 20 04:00:21.243: %OSPF-6-DFT_OPT: Protocol timers for fast convergence are Enabled.
R1(config-router)#network 200.0.0.0 0.0.0.3 area 0
R1(config-router)#network 192.168.1.0 0.0.0.255 area 0
R1(config-router)#network 192.168.2.0 0.0.0.255 area 0
R1(config-router)#router-id 1.1.1.1
R1(config-router)#ex
*May 20 04:01:14.653: %OSPF-6-NEW_RTRID: New router-id will take effect immediately. If OSPF adjacencies are UP for OSPF-1 they will be reset.
R1(config-router)#ex
```

FIRST ISSUE FOUND!

Little explanation:

I decided to see OSPF neighbors from router 2 with command #do show ip ospf neighbor, but for some reason nothing was coming up I also checked if OSPF was properly configured.

```
R2(config)#do show ip ospf neighbor
R2(config)#do show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance      Last Update
```

I found the solution by running the configuration on R2, everything was ok but on Router one for some reason the interfaces were not properly configured.

```
interface Ethernet0/0
 ip address 200.0.0.1 255.255.255.252
!
interface Ethernet0/1
 no ip address
 shutdown
!
interface Ethernet0/2
 ip address 192.168.2.1 255.255.255.0
!
```

Now lets go ahead and properly configured interfaces. They were able to form adjacencies.

```
R1(config)#int eth0/0
R1(config-if)#no ip address 200.0.0.1 255.255.255.252
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#int eth0/1
R1(config-if)#ip address 200.0.0.1 255.255.255.252
R1(config-if)#no shut
R1(config-if)#exi
R1(config)#do show ip int
*May 20 18:00:29.232: %LINK-5-UPDOWN: Interface Ethernet0/1, changed state to up
*May 20 18:00:30.233: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to up
R1(config)#do show ip inte brief
*May 20 18:00:37.461: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Ethernet0/1 from LOADING to FULL, Loading Done
```

We are finally able to see the OSPF neighbor, as you can see the R2 is elected as the DR because it has a higher router ID configured but can we change it ? LET'S PROBE IT.

```
R1(config)#do show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
2.2.2.2          1    FULL/DR         00:00:37   200.0.0.2      Ethernet0/1
R1(config)#do show ip ospf int eth0/1
Ethernet0/1 is up, line protocol is up
  Internet Address 200.0.0.1/30, Interface ID 3, Area 0
  Attached via Network Statement
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 10
  Topology-MTID     Cost    Disabled    Shutdown    Topology Name
    0                10        no          no          Base
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 2.2.2.2, Interface address 200.0.0.2
  Backup Designated router (ID) 1.1.1.1, Interface address 200.0.0.1
```

I changed the router ID on R1.

```
R1(config)#router ospf 1
R1(config-router)#router-id 3.3.3.3
R1(config-router)#
*May 20 18:05:27.960: %OSPF-6-NEW_RTRID: New router-id will take effect immediately.
```

But R1 still BDR.

```
R2(config)#do show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
3.3.3.3          1    FULL/BDR         00:00:37   200.0.0.1      Ethernet0/0
```

LEST TRY ONE MORE THING. IMPORTANT IN REAL ENVIRONMENT THIS CAN CAUSE CHAOS!!!!!!! Clear IP OSPF process.

```

R1#clear ip ospf process
Reset ALL OSPF processes? [no]: yes
R1#
*May 20 18:08:20.930: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Ethernet0/1 from FULL to INIT
*May 20 18:08:20.934: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Ethernet0/1 from INIT to DOWN
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#do show ip ospf neighbor

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	FULL/DR	00:00:31	200.0.0.2	Ethernet0/1

REMEMBER ONE THING OSPF IS NOT PREEMPTIVE therefore it maintains router 2 as the DR. However, if we try one last thing. Clear OSPF process on router 2.

```

R2#clear ip ospf process
Reset ALL OSPF processes? [no]: yes
R2#
*May 20 18:11:10.817: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Ethernet0/0 from FULL to INIT
*May 20 18:11:10.821: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Ethernet0/0 from LOADING to DOWN
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#do show ip ospf neighbor

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
3.3.3.3	1	FULL/DR	00:00:38	200.0.0.1	Ethernet0/0

```

R1(config)#do show ip ospf neighbor

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	0	FULL/-	00:00:35	200.0.0.2	Ethernet0/1

As you can see in the picture, there is no reason to have a DR or BDR election, since this is a point-to-point network type.

PING FROM PC2 TO SERVER.

```

inserthostname-here:~$ ping 10.0.0.4
PING 10.0.0.4 (10.0.0.4): 56 data bytes
64 bytes from 10.0.0.4: seq=0 ttl=42 time=4.996 ms
64 bytes from 10.0.0.4: seq=1 ttl=42 time=1.355 ms
^V64 bytes from 10.0.0.4: seq=2 ttl=42 time=1.927 ms
c64 bytes from 10.0.0.4: seq=3 ttl=42 time=3.201 ms
^C
--- 10.0.0.4 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 1.355/2.869/4.996 ms

```

PING FROM PC1 TO SERVER. It did not work, I also tried to use traceroute to see where the packet goes, to be able to see the hops. It never leaves its own interface.

```

PING 10.0.0.4 (10.0.0.4): 56 data bytes
^C
--- 10.0.0.4 ping statistics ---
5 packets transmitted, 0 packets received, 100% packet loss
inserthostname-here:~$ traceroute 10.0.0.4
traceroute to 10.0.0.4 (10.0.0.4), 30 hops max, 46 byte packets
 1  192.168.1.4 (192.168.1.4)  3112.198 ms !H  3148.993 ms !H  3117.523 ms !H

```

I decided to see the interface configuration, because it was so weird that it did not tried to reach its default gateway, which is 192.168.1.1

```
R1(config)#do show ip int brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	192.168.10.1	YES	manual	up	up
Ethernet0/1	200.0.0.1	YES	manual	up	up
Ethernet0/2	192.168.2.1	YES	NVRAM	up	up

As you can see the wrong address was configured on the interface instead of 192.168.1.1/24 the IP Address is 192.168.10.1, lets change it.

```
R1(config)#int eth0/0
R1(config-if)#no ip address 192.168.10.1 255.255.255.0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exi
R1(config)#do wr
Building configuration...
```

There you go, the ping works.

```
inserthostname-here:~$ ping 10.0.0.4
PING 10.0.0.4 (10.0.0.4): 56 data bytes
64 bytes from 10.0.0.4: seq=0 ttl=42 time=3.351 ms
64 bytes from 10.0.0.4: seq=1 ttl=42 time=2.890 ms
^C
--- 10.0.0.4 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 2.890/3.120/3.351 ms
```